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JupyterLab Python [conda env:base] * Anaconda Toolbox

```
[21]: import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
df=sns.load_dataset("titanic")
print(df)
#Discriptive Statistics
print("Discriptive Statistics")
print(df.describe)
#Skewed
numeric_cols=df.select_dtypes(include=np.number)
print(numeric_cols.skew())
#Z_score
zscore_df=numeric_cols.copy()
for col in zscore_df.columns:
    zscore_df[col]=(
        zscore_df[col]-zscore_df[col].mean()
    )/zscore_df[col].std()
print(zscore_df.head())
print(zscore_df.mean())
print(zscore_df.std())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	
..	...	...	...	...	...	...	...	...	...	

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JupyterLab Python [conda env:base] * Anaconda Toolbox

```
parch      1.0
fare       1.0
dtype: float64

[22]: #pearson
pearson_corr=numeric_cols.corr(method='pearson')
print(pearson_corr)
#heatMap
plt.figure(figsize=(8,5))
sns.heatmap(
    pearson_corr,
    annot=True,
    cmap='coolwarm'
)
plt.title("Pearson Correlation")
plt.show()
```

	survived	pclass	age	sibsp	parch	fare
survived	1.000000	-0.338481	-0.077221	-0.035322	0.081629	0.257307
pclass	-0.338481	1.000000	-0.369226	0.083081	0.018443	-0.549500
age	-0.077221	-0.369226	1.000000	-0.308247	-0.189119	0.096067
sibsp	-0.035322	0.083081	-0.308247	1.000000	0.414838	0.159651
parch	0.081629	0.018443	-0.189119	0.414838	1.000000	0.216225
fare	0.257307	-0.549500	0.096067	0.159651	0.216225	1.000000

Pearson Correlation

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Code

Python [conda env:base] * Anaconda Toolbox

parc

0.082 0.018 -0.19 0.41 1 0.22

fare

0.26 -0.55 0.096 0.16 0.22 1

survived pclass age sibsp parch fare

[23]:

#Spearman Correlation
Spearman_corr=numeric_cols.corr(method='spearman')
print(Spearman_corr)
plt.figure(figsize=(8,5))
sns.heatmap(
 pearson_corr,
 annot=True,
 cmap='viridis'
)
plt.title("Spearman Correlation")
plt.show()

survived pclass age sibsp parch fare

survived 1.000000 -0.339668 -0.052565 0.088879 0.138266 0.323736

pclass -0.339668 1.000000 -0.361666 -0.043019 -0.022801 -0.688032

age -0.052565 -0.361666 1.000000 -0.182061 -0.254212 0.135051

sibsp 0.088879 -0.043019 -0.182061 1.000000 0.450014 0.447113

parch 0.138266 -0.022801 -0.254212 0.450014 1.000000 0.410074

fare 0.323736 -0.688032 0.135051 0.447113 0.410074 1.000000

Spearman Correlation

1 -0.34 -0.077 -0.035 0.082 0.26

survived

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parch	0.082	0.018	-0.19	0.41	1	0.22
fare	0.26	-0.55	0.096	0.16	0.22	1
	survived	pclass	age	sibsp	parch	fare

[25]:

#Compare Spearman & pearson
comparison = pd.DataFrame({
 "Pearson":pearson_corr["age"],
 "Spearman":Spearman_corr["age"]
})
print(comparison)

	Pearson	Spearman
survived	-0.077221	-0.052565
pclass	-0.369226	-0.361666
age	1.000000	1.000000
sibsp	-0.308247	-0.182061
parch	-0.189119	-0.254212
fare	0.096067	0.135051

[]:

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```
[3]: import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
df=sns.load_dataset("titanic")
print(df)
print(df.head())
print(df.shape)
df.info()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	
..	...	...	...	...	...	...	...	...	...	
886	0	2	male	27.0	0	0	13.0000	S	Second	
887	1	1	female	19.0	0	0	30.0000	S	First	
888	0	3	female	NaN	1	2	23.4500	S	Third	
889	1	1	male	26.0	0	0	30.0000	C	First	
890	0	3	male	32.0	0	0	7.7500	Q	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True
..	...	...	...	...	...	...
886	man	True	NaN	Southampton	no	True
887	woman	False	B	Southampton	yes	True

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Python [conda env:base]

memory usage: 80.7+ KB

[6]:
#Inconsistent Formats
df.loc[0, 'embark_town']='SOUTHAMPTON'
df.loc[1, 'embark_town']='SouthAmpton'
df.loc[2, 'embark_town']='southampton'
print(df['embark_town'].unique())

['SOUTHAMPTON' 'SouthAmpton' 'southampton' 'Southampton' 'Queenstown'
 'Cherbourg' nan]

[7]:
#Capitalize the First Letter
df['embark_town']=df['embark_town'].str.title()
print(df['embark_town'].unique())

['Southampton' 'Queenstown' 'Cherbourg' nan]

[8]:
print(df['embark_town'].isnull().sum())

2

[9]:
df['embark_town']=df['embark_town'].fillna('Kingstown')
print(df['embark_town'].unique())

['Southampton' 'Queenstown' 'Cherbourg' 'Kingstown']

[23]:
#Duplicate rows
dupi_rows=df.iloc[:5]
df_dup=pd.concat([df,dupi_rows])
print(df_dup.shape)
print(df_dup.duplicated().sum())
df_dup[df_dup.duplicated()]

(896, 15)
112

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JupyterLab Python [conda env:base] * Anaconda Toolbox

3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

112 rows × 15 columns

[25]: print(df_dup.duplicated().sum())

112

[26]: df_dup[df_dup.duplicated()]

[26]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
47	1	3	female	NaN	0	0	7.7500	Q	Third	woman	False	NaN	Queenstown	yes	True
76	0	3	male	NaN	0	0	7.8958	S	Third	man	True	NaN	Southampton	no	True
77	0	3	male	NaN	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True
87	0	3	male	NaN	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True
95	0	3	male	NaN	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	True
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Southampton	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	no	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

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JupyterLab Python [conda env:base] * Anaconda Toolbox

[28]: #Remove Duplicates

df_dup=df_dup.drop_duplicates()

print(df_dup.duplicated().sum())

0

[29]: #Spelling Mistakes

df.loc[0,'embark_town']='Southampton'

df.loc[1,'embark_town']='Southhhampton'

print(df['embark_town'].value_counts())

embark_town

Southampton 644

Cherbourg 167

Queenstown 77

Kingstown 2

Southhhampton 1

Name: count, dtype: int64

[30]: #Replace

df['embark_town']=df['embark_town'].replace({

'Southampton':'Southampton',

'Southhhampton':'Southhhampton'

})

print(df['embark_town'].value_counts())

embark_town

Southampton 644

Cherbourg 167

Queenstown 77

Kingstown 2

Southhhampton 1

Name: count, dtype: int64

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JupyterLab Python [conda env:base] * Anaconda Toolbox

Kingstown 2

Southhampton 1

Name: count, dtype: int64

[31]: #Detect possible types

print(sorted(df['embark_town'].dropna().unique()))

['Cherbourg', 'Kingstown', 'Queenstown', 'Southampton', 'Southhampton']

[32]: #Unit mismatches

df.loc[0,'age']=240

df.loc[1,'age']=300

df['age'].describe()

[32]: count 714.000000

mean 30.371387

std 19.362012

min 0.420000

25% 20.125000

50% 28.000000

75% 38.750000

max 300.000000

Name: age, dtype: float64

[33]: #Fix unit mismatch

df.loc[df['age']>120,'age']=(

df.loc[df['age']>120,'age']/12

)

df['age'].describe()

[33]: count 714.000000

mean 29.678109

std 14.525902

min 0.420000

25% 20.000000

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JupyterLab Python [conda env:base]

```
max 88.000000
Name: age, dtype: float64

[34]: #wrong Category Labels
# df['survived']
#contains
#0,1,not very readable
df['survived_label']=df['survived'].map({
    0:'No',
    1:'Yes'
})
print(df.head())

   survived  pclass    sex  age  sibsp  parch    fare embarked  class \
0         0       3  male  20.0     1     0   7.2500      S  Third
1         1       1  female  25.0     1     0  71.2833      C  First
2         1       3  female  26.0     0     0   7.9250      S  Third
3         1       1  female  35.0     1     0  53.1000      S  First
4         0       3  male   35.0     0     0   8.0500      S  Third

   who  adult_male  deck  embark_town  alive  alone  survived_label
0   man         True  NaN  Southampton    no   False              No
1 woman        False   C  Southhampton  yes  False              Yes
2 woman        False  NaN  Southampton  yes   True              Yes
3 woman        False   C  Southampton  yes  False              Yes
4   man         True  NaN  Southampton    no   True               No

[34]: #wrong Category Labels
# df['survived']
#contains
#0,1,not very readable
df['survived_label']=df['survived'].map({
    0:'No',
    1:'Yes'
})
```

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Python [conda env:base] * Anaconda Toolbox

```
[40]: df.drop(columns=['deck'],inplace=True)
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	embark_town	alive	alone	survived_lable
0	0	3	male	20.0	1	0	7.2500	S	Third	man	True	Southampton	no	False	No
1	1	1	female	25.0	1	0	71.2833	C	First	woman	False	Southhhampton	yes	False	Yes
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	Southampton	yes	True	Yes
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	Southampton	yes	False	Yes
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	Southampton	no	True	No

```
[ ]: #Remove
fd.drop(columns=['deck'],inplace=True df.head())
```

```
[37]: #Handling null values
# Null values occur when data is unavailable unknown or not recorded
#Check miissing values
df.isnull().sum()
```

survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0

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Taskbar icons: File Explorer, Edge, VS Code, JupyterLab, etc.

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The screenshot shows the Anaconda Toolbox application interface. At the top, there is a header bar with the Anaconda logo, the text "Anaconda Toolbox", and icons for a home button, a cloud, and a code editor. The version number "v4.20.0" is displayed on the right. Below the header, there is a large green banner with the Anaconda logo and the text "Anaconda Your Toolbox for Python Projects". Underneath the banner, there is a section titled "Anaconda Cloud" with a cloud icon. This section contains three buttons: "Create a New Project", "Create a New Notebook", and "My Projects", each with a right-pointing arrow. Below this, there is a section titled "Code Snippets" with a code icon. It contains one button: "Manage Code Snippets" with a right-pointing arrow. At the bottom, there is a section titled "Anaconda AI Assistant" with a star icon. It contains one button: "Chat-based Python Help" with a right-pointing arrow.

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File Edit View Run Kernel Settings Help

+
-
Copy
Paste
Run
Stop
Code

Python [conda env:base] *
Anaconda Toolbox

```

embark_town      0
alive            0
alone            0
survived_lable   0
dtype: int64

[41]: #Replace missing values
df['age'] = df['age'].fillna(df['age'].mean())
df.head(20)

[41]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	embark_town	alive	alone	survived_lable
0	0	3	male	20.000000	1	0	7.2500	S	Third	man	True	Southampton	no	False	No
1	1	1	female	25.000000	1	0	71.2833	C	First	woman	False	Southhampton	yes	False	Yes
2	1	3	female	26.000000	0	0	7.9250	S	Third	woman	False	Southampton	yes	True	Yes
3	1	1	female	35.000000	1	0	53.1000	S	First	woman	False	Southampton	yes	True	Yes
4	0	3	male	35.000000	0	0	8.0500	S	Third	man	True	Southampton	no	False	No
5	0	3	male	29.678109	0	0	8.4583	Q	Third	man	True	Southampton	no	False	No
6	0	1	male	54.000000	0	0	51.8625	S	First	man	True	Southampton	no	False	No
7	0	3	male	2.000000	3	1	21.0750	S	Third	child	False	Southampton	yes	True	Yes
8	1	3	female	27.000000	0	2	11.1333	S	Third	woman	False	Southampton	yes	True	Yes
9	1	2	female	14.000000	1	0	30.0708	C	Second	child	False	Southampton	yes	True	Yes
10	1	3	female	4.000000	1	1	16.7000	S	Third	child	False	Southampton	yes	True	Yes
11	1	1	female	58.000000	0	0	26.5500	S	First	woman	False	Southampton	yes	True	Yes
12	0	3	male	20.000000	0	0	8.0500	S	Third	man	True	Southampton	no	False	No

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18	0	3	female	31.000000	1	0	18.0000	S	Third	woman	False	Southampton	no	False	No
19	1	3	female	29.678109	0	0	7.2250	C	Third	woman	False	Cherbourg	yes	True	Yes

[43]:

```
#Median imputation
#Replace missing values with middle values
#Lets check age distribution
#df['age'].skew()
#if skew is high:use median
#if data is nearly normal:use mean

df['age']=df['age'].fillna(df['age'].median())
df.head(20)
#Replace missng values with most frequent values
#df['embarked_mode']=df['embarked_mode'].fillna(df['embarked'].mode()[0])
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male
0	0	3	male	20.000000	1	0	7.2500	S	Third	man	True
1	1	1	female	25.000000	1	0	71.2833	C	First	woman	False
2	1	3	female	26.000000	0	0	7.9250	S	Third	woman	False
3	1	1	female	35.000000	1	0	53.1000	S	First	woman	False
4	0	3	male	35.000000	0	0	8.0500	S	Third	man	True
5	0	3	male	29.678109	0	0	8.4583	Q	Third	man	True
6	0	1	male	54.000000	0	0	51.8625	S	First	man	True
7	0	3	male	2.000000	3	1	21.0750	S	Third	child	False
8	1	3	female	27.000000	0	2	11.1333	S	Third	woman	False

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19	1	3	female	29.678109	0	0	7.2250	C	Third	woman	False	Cherbourg	yes	True	Yes
----	---	---	--------	-----------	---	---	--------	---	-------	-------	-------	-----------	-----	------	-----

[44]:

#Forward fill
print(df['embarked'].isnull().sum())

2

[45]:

#Apply forward fill
df['embarked_ffill']=df['embarked'].ffill()
#Check missing values
print(df['embarked'].isnull().sum())

2

[50]:

#KNN imputation
#K-neraest neighbours imputation fills missing values using information from similar rows
#instead of using single values(mean/median),it finds the k most similar records and uses their
import pandas as pd
import numpy as np
data={
 'age': [22,35,45,56,np.nan,85],
 'fare': [7.25,8.05,7.90,7.50,8.00,9.00]
}
df=pd.DataFrame(data)
print(df)

 age fare
0 22.0 7.25
1 35.0 8.05
2 45.0 7.90
3 56.0 7.50
4 NaN 8.00
5 85.0 9.00

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3 50.0 7.50

4 NaN 8.00

5 85.0 9.00

[52]:

#Apply KNN imputation
from sklearn.impute import KNNImputer
KNN=KNNImputer(n_neighbors=3)
df_KNN=pd.DataFrame(
 KNN.fit_transform(df),
 columns=df.columns)
print(df_KNN)

age fare

0 22.000000 7.25

1 35.000000 8.05

2 45.000000 7.90

3 56.000000 7.50

4 45.333333 8.00

5 85.000000 9.00

[53]:

#SKLearn->scikitLearn library for machine learning and data preprocessing
#impute-> Module for handling missing values
#KNNImputer->Class that fills missing values using K-Nearest
#fit-transform(df)-> learns from data and fills missing values
#converts the result back into a pandas dataframe

[56]:

#Check missing values
import pandas as pd
import seaborn as sns
from sklearn.impute import KNNImputer
#Load dataset
df=sns.load_dataset('titanic')
df['embarked_ffill']=df['embarked'].ffill
cols=[
 'age',

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#KNNImputer->Class that fills missing values using K-Nearest

#fit-transform(df)-> learns from data and fills missing values

#converts the result back into a pandas dataframe

[56]:

#Check missing values

import pandas as pd

import seaborn as sns

from sklearn.impute import KNNImputer

#Load dataset

df=sns.load_dataset('titanic')

df['embarked_ffill']=df['embarked'].ffill

cols=[

'age',

'fare',

'sibsp',

'parch',

'pclass']

imputer=KNNImputer(n_neighbors=5)

df[cols]=imputer.fit_transform(df[cols])

print(df.isnull().sum())

survived

0

pclass

0

sex

0

age

0

sibsp

0

parch

0

fare

0

embarked

2

class

0

who

0

adult_male

0

deck

688

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alone 0

embarked_ffill 0

dtype: int64

[57]:

#Bcakward fill
#df.fillna(method='bfill',inplace=True) or df.bfill(inplace=True)
#Encoding
#convert the 6text into numerical values for the understandable
#Eg male,femake a model cannot directly understand male ,female ,we convert catogries into numbers using encoding
#Types of encoding
#1.Label encoding(ordinal data)
#Assign number toneach category
#eg: small,medium,large becomes size 0,1,2
#Ordinal (small<medium<large),Nominal (Male,female)
#2.One Hot encoding
#Creates a separate binary column for each category
#Eg:(Nominal Data eg:country,color,gender,city,department)
#3.Target Encoding
#Replace categories with the average value of the target variable
#Eg (imagine 5000 cities one-hot encoding would create 5000 columns Not practical but Target en

[62]:

import pandas as pd
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
df=sns.load_dataset('titanic')
le=LabelEncoder()
df['sex_encoded']=le.fit_transform(
df['sex']
)
df=pd.get_dummies(
df,
columns=['embarked'],
drop_first=True

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 Code
 

```
[62]: import pandas as pd
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
df=sns.load_dataset('titanic')
le=LabelEncoder()
df['sex_encoded']=le.fit_transform(
    df['sex']
)
df=pd.get_dummies(
    df,
    columns=['embarked'],
    drop_first=True
)
target_map=df.groupby(
    'pclass'
)['survived'].mean()
df['class_target']=(
    df['pclass']
    .map(target_map)
)
print(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	class	who
0	0	3	male	22.0	1	0	7.2500	Third	man
1	1	1	female	38.0	1	0	71.2833	First	woman
2	1	3	female	26.0	0	0	7.9250	Third	woman
3	1	1	female	35.0	1	0	53.1000	First	woman
4	0	3	male	35.0	0	0	8.0500	Third	man

	adult_male	deck	embark_town	alive	alone	sex_encoded	embarked_Q
0	True	NaN	Southampton	no	False	1	False
1	False	C	Cherbourg	yes	False	0	False
2	False	NaN	Southampton	yes	True	0	False

A screenshot of a Windows desktop environment. In the background, a Visual Studio Code window is open, displaying a C# script named 'NewGame.cs'. The code includes comments in Chinese and defines a 'NewGame' method. A sidebar on the left shows the project explorer with folders like 'Assets' and 'Scripts'. Overlaid on top of the code editor is the 'Snipping Tool' application window. It has a title bar 'Snipping Tool' and a close button. The main area of the tool contains the text 'Screenshot copied to clipboard' and 'Automatically saved to screenshots folder.' Below this text are two buttons: 'Markup and share' and another button that is partially obscured.

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```
3 True 0.629630
4 True 0.242363
```

[65]:
from sklearn.preprocessing import MinMaxScaler
scaler=MinMaxScaler()
df[['age_minmax','fare_minmax']]=scaler.fit_transform(
 df[['age','fare']]
)
df[['age_minmax','fare_minmax']].head()

[65]:
age_minmax fare_minmax
0 0.271174 0.014151
1 0.472229 0.139136
2 0.321438 0.015469
3 0.434531 0.103644
4 0.434531 0.015713

[68]:
import matplotlib.pyplot as plt
plt.hist(df['fare'].dropna(),bins=30)
plt.title("Original fare")
plt.show()
plt.hist(df['fare_minmax'],bins=30)
plt.title("Minmax Scaled fare")
plt.show()

Original fare

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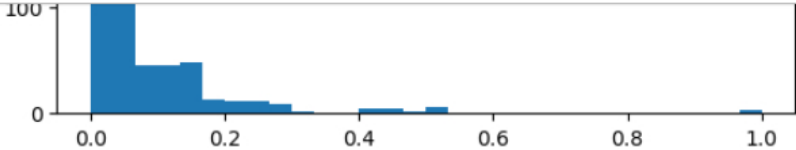
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A histogram showing the distribution of 'fare_std' values. The x-axis ranges from 0.0 to 1.0, and the y-axis ranges from 0 to 100. The distribution is highly right-skewed, with a peak at 0.0 and a long tail extending towards 1.0.

[70]:

```
from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
df[['age_std','fare_std']]=scaler.fit_transform(
    df[['age','fare']]
)
df[['age_std','fare_std']].head()
```

[70]:

	age_std	fare_std
0	-0.530377	-0.502445
1	0.571831	0.786845
2	-0.254825	-0.488854
3	0.365167	0.420730
4	0.365167	-0.486337

[71]:

```
from sklearn.preprocessing import RobustScaler
scaler=RobustScaler()
df[['age_std','fare_std']]=scaler.fit_transform(
    df[['age','fare']]
)
df[['age_std','fare_std']].head()
```

[71]:

	age_std	fare_std
--	---------	----------

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